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Single Particle Detection

Single-shot Self-supervised Particle Detection in Microscopy

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Outline

Status Update

Project Overview

- **Goal**: Implement and evaluate LodeSTAR for particle detection
- **Scope**: Single-particle detection on synthetic microscopy images
- **Particle Types**: Janus, Ring, Spot, Ellipse, Rod
- **Implementation**: PyTorch + DeepTrack + PyTorch Lightning
- **Tracking**: Logger and Weights & Biases (WandB) for experiment monitoring

Data Generation Pipeline

Image Generation:

- **Synthetic particles** with realistic parameters
- **Multiple datasets** for different scenarios:
  - Same shape, same size
  - Same shape, different size
  - Different shape, same size
  - Different shape, different size
- **XML annotations** with bounding boxes and SNR values

Technical Details:

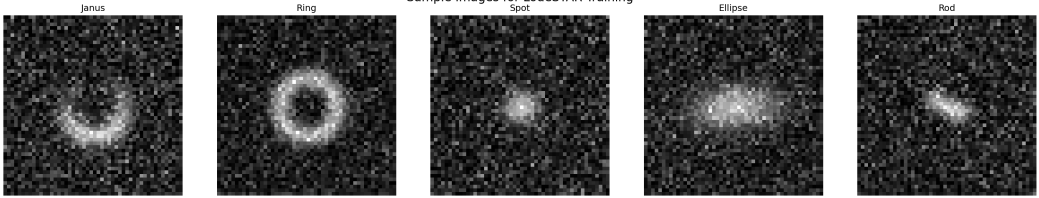
- **Image size**: 416×416 pixels
- **SNR range**: 1-30
- **Intensity range**: 0.1-1.0
- **Particle parameters**: Type-specific values
- **100 images** per dataset type

Sample Images Generation

Sample Images:

- **Clean samples** for each particle type
- **50 $\times$ 50 pixels** for training
- **High SNR** (10) for clear features
- **Realistic parameters** for each particle type
- **Stored in** `data/Samples/`

Sample Images for LodeSTAR Training



Training Pipeline Implementation

Training Setup:

- **Single-particle models** for each particle type
- **Data augmentation**: Random multiplication and addition
- **400 samples per epoch** (configurable)
- **200 epochs** maximum training
- **Batch size**: 8 (balanced for memory)

Technical Stack:

- **PyTorch Lightning** for training
- **DeepTrack** for data pipelines
- **WandB** for experiment tracking
- **Mixed precision** training (16-bit)
- **GPU acceleration** when available

Training Configuration & Optimization

Parameter	Value
Learning Rate	0.0001
Batch Size	8
Max Epochs	200
N Transforms	4
Alpha	1.0
Beta	0.0
Cutoff	0.8

Experiment 192 Results: 6-cycle debugging optimized parameters

- **Default LodeSTAR** architecture performs best (F1: 0.526)
- **Final configuration**: 50px images, n_transforms=4, only intensity augmentation

Testing Pipeline & Results

Testing Setup:

- **Load trained models** from checkpoints
- **Test on generated datasets** in `data/Testing/`
- **Parse XML annotations** for ground truth
- **Calculate detection metrics**: Precision, Recall, F1
- **Distance threshold**: 20 pixels for positive detection

Training Success:

- **All 5 models** completed 200 epochs
- **Loss convergence** observed
- **Checkpoints saved** every epoch
- **WandB logging** active
- **Model weights** exported

Results & Discussion

Current Status & Key Results

Completed:

- **All 5 models** successfully trained (200 epochs each)
- **Comprehensive evaluation** across 4 dataset types
- **Architecture debugging** completed (Experiment 192)
- **Performance analysis** with statistical metrics

Key Findings:

- **Default LodeSTAR** architecture performs best
- **High recall** on same shape scenarios (0.89-0.97)
- **Moderate performance** on different shapes
- **Optimized parameters** identified

Performance Results Summary

Dataset Type	Avg F1	Avg Precision	Avg Recall	Best Model
SSSS	0.561	0.638	0.919	Ring (0.800)
SSDS	0.394	0.488	0.889	Rod (0.524)
DSSS	0.254	0.534	0.439	Janus (0.273)
DSDS	0.261	0.399	0.625	Spot (0.397)

Key Insights:

- **Best performance** on same shape scenarios (expected)
- **Ring model** excels on same shape (F1: 0.800)
- **Rod model** shows good generalization
- **Lower performance** on different shapes (expected)

Model Performance Summary

Top Performers:

- **Ring**: F1: 0.800 (same shape)
- **Rod**: F1: 0.524 (good generalization)
- **Spot**: F1: 0.670 (consistent)
- **Janus**: F1: 0.469 (moderate)
- **Ellipse**: F1: 0.150 (struggles)

Performance Patterns:

- **Same shape**: High recall (0.89-0.97)
- **Different shape**: Lower recall (0.44-0.63)
- **Precision varies**: 0.24-0.89 across scenarios
- **Shape-specific** performance differences

Key Findings & Future Work

Key Achievements:

- **Self-supervised learning** works for particle detection
- **High recall** on same shape scenarios (0.89-0.97)
- **Default LodeSTAR** architecture performs best
- **Complete pipeline** implemented successfully

Areas for Improvement:

- **Low precision** on different shapes
- **High false positive rates**
- **Shape-specific training** needed
- **Parameter optimization** required